

Denial Prediction POC

Version 1.5 — Proposed Feature Engineering Strategy

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Subject	Proposed feature engineering strategy for Denial Prediction POC — Version 1.5
Focus	Transition from identity-driven prediction to behaviour-driven prediction

1. Objective

The objective of Version 1.5 is to transition from an identity-driven historical prediction model to a behaviour-driven prediction model.

The ML model will not replicate Thynk's deterministic rule engine. Instead, it will learn generic claim behaviours from historical data that can generalize to new payers, TPAs, hospitals, and providers.

2. Proposed Features for Version 1.5

The table below sets out the proposed feature set for Version 1.5, organized by behaviour category, together with the behaviour each feature captures and the rationale for selection.

Feature Category	Feature	Behaviour Captured	Why Selected
Patient Behaviour	Patient Age	Patient demographic behaviour	Different age groups exhibit different claim complexity and utilization patterns.
Patient Behaviour	Gender	Patient demographic behaviour	Certain clinical services naturally vary by gender.
Clinical Behaviour	Diagnosis Count (Primary + Secondary)	Clinical complexity	Claims with multiple diagnoses generally indicate higher complexity and documentation requirements.
Clinical Behaviour	Activity Quantity	Treatment intensity	Higher service utilization often represents more complex encounters.
Financial Behaviour	Activity Gross	Financial behaviour	Represents overall billed value of the service.
Financial Behaviour	Unit Cost	Cost behaviour	Identifies expensive procedures independent of quantity.
Financial Behaviour	Discount Ratio	Billing behaviour	Captures pricing adjustments and billing patterns.
Financial Behaviour	High Cost Indicator	Financial risk behaviour	Flags unusually high-cost claims that may require additional review.

Feature Category	Feature	Behaviour Captured	Why Selected
Financial Behaviour	IP Stay to Cost Ratio	Resource utilization behaviour	Measures relationship between inpatient stay and billed amount.
Operational Behaviour	Encounter Type	Operational workflow	Inpatient, outpatient and emergency encounters follow different billing workflows.
Operational Behaviour	Length of Stay	Clinical severity	Longer stays generally indicate more complex treatment episodes.
Operational Behaviour	Billing Lag Days	Operational efficiency	Delayed claim submission may indicate documentation or workflow issues.
Insurance Behaviour	Insurance Plan Tier	Benefit behaviour	Different benefit plans have varying authorization requirements and coverage limits.
Insurance Behaviour	Network	Benefit behaviour	Network determines covered services, authorization requirements and member benefits.
Provider Behaviour	Profession	Clinical practice behaviour	Different provider specialties naturally exhibit different treatment patterns.
Provider Behaviour	Category	Provider behaviour	Captures provider classification without relying on individual identity.
Provider Behaviour	Facility Type	Organization behaviour	Hospital, clinic and day-care facilities exhibit different operational workflows.

3. Features Removed from Version 1.0

The following identity-based and historical-aggregate features from Version 1.0 have been removed in Version 1.5, as they limit generalization to new entities.

Removed Feature	Reason for Removal
payer_id	Entity identifier. Causes model memorization and poor generalization to new payers.
receiver_id	Entity identifier. Does not represent claim behaviour.
tpa_id	Entity identifier. Creates cold-start problems for new TPAs.
clinician_license / clinician_id	Learns individual providers rather than provider behaviour.
facility_license / facility_id	Learns specific organizations instead of facility characteristics.
medical_necessity_score	Historical aggregate score. Represents previous outcomes rather than intrinsic claim behaviour.
coverage_score	Historical payer-specific statistic. Strong dependency on payer identity.

Removed Feature	Reason for Removal
pa_risk_score	Historical authorization metric rather than claim behaviour.
clinician_success_score	Learns historical clinician performance instead of provider characteristics.
facility_success_score	Learns historical facility performance instead of organizational behaviour.
Exact Primary Diagnosis Code	Encourages memorization of diagnosis identities instead of learning generic clinical complexity.
Exact Activity (CPT) Code	Highly specific feature with limited generalization to unseen procedures.
Exact ICD-CPT Pair	Represents historical combinations rather than generalized claim behaviour.

4. SME Suggested Features (Berlin Discussion)

Feature suggestions raised during the Berlin SME discussion were evaluated against the Version 1.5 generalization criteria, with the following outcomes.

Suggested Feature	Decision	Included	Justification
Insurance Plan Tier	Accepted	Yes	Represents benefit structure and authorization behaviour. Generic across insurers.
Network	Accepted (if available in source data)	Yes	Represents plan benefits, authorization requirements and member coverage. Behavioural feature.
TPA ID	Rejected	No	Identity-based feature. Does not generalize to new TPAs.
Receiver ID	Rejected	No	Identity-based feature. Does not describe claim behaviour.

5. Behaviour vs Identity Philosophy

Each identity-based feature from Version 1.0 has been replaced with one or more behaviour-based features that capture the same underlying signal without relying on entity identity.

Identity-Based Feature	Behaviour-Based Replacement
Payer ID	Insurance Plan Tier + Network
Facility ID	Facility Type
Clinician ID	Profession + Category
Historical Success Scores	Financial, Operational and Clinical Behaviour Features
Exact Diagnosis Codes	Diagnosis Count / Clinical Complexity
Exact Procedure Codes	Activity Quantity / Financial Characteristics

6. Feature Engineering Philosophy

The Version 1.5 model is designed around the following behavioural dimensions:

Behaviour Dimension	Purpose
Patient Behaviour	Understand demographic influences without relying on patient identity.
Clinical Behaviour	Measure clinical complexity rather than diagnosis identities.
Financial Behaviour	Learn billing and utilization patterns associated with denials.
Operational Behaviour	Capture workflow characteristics that influence claim quality.
Insurance Behaviour	Learn benefit-plan characteristics instead of payer identities.
Provider Behaviour	Learn treatment patterns based on provider type rather than individual clinicians.

7. Alignment with Product Vision

This feature engineering approach aligns with the agreed product direction:

- ✓ Thynk continues to handle deterministic business rules (eligibility, authorization, medical necessity, coverage, etc.).
- ✓ Machine Learning focuses only on learning historical behavioural patterns that are not explicitly captured by rule engines.
- ✓ Features are designed to generalize across new payers, TPAs, hospitals and providers without relying on entity identifiers.
- ✓ Every selected feature has a clear business interpretation and can be explained to operational stakeholders.

8. Final Version 1.5 Behaviour Framework

The table below consolidates the final feature set by behaviour pillar.

Behaviour Pillar	Features
Patient Behaviour	Patient Age, Gender
Clinical Behaviour	Diagnosis Count (Primary + Secondary), Activity Quantity
Financial Behaviour	Activity Gross, Unit Cost, Discount Ratio, High Cost Indicator, IP Stay-to-Cost Ratio
Operational Behaviour	Encounter Type, Length of Stay, Billing Lag Days
Insurance Behaviour	Insurance Plan Tier, Network (if available)
Provider Behaviour	Profession, Category, Facility Type

End of report.